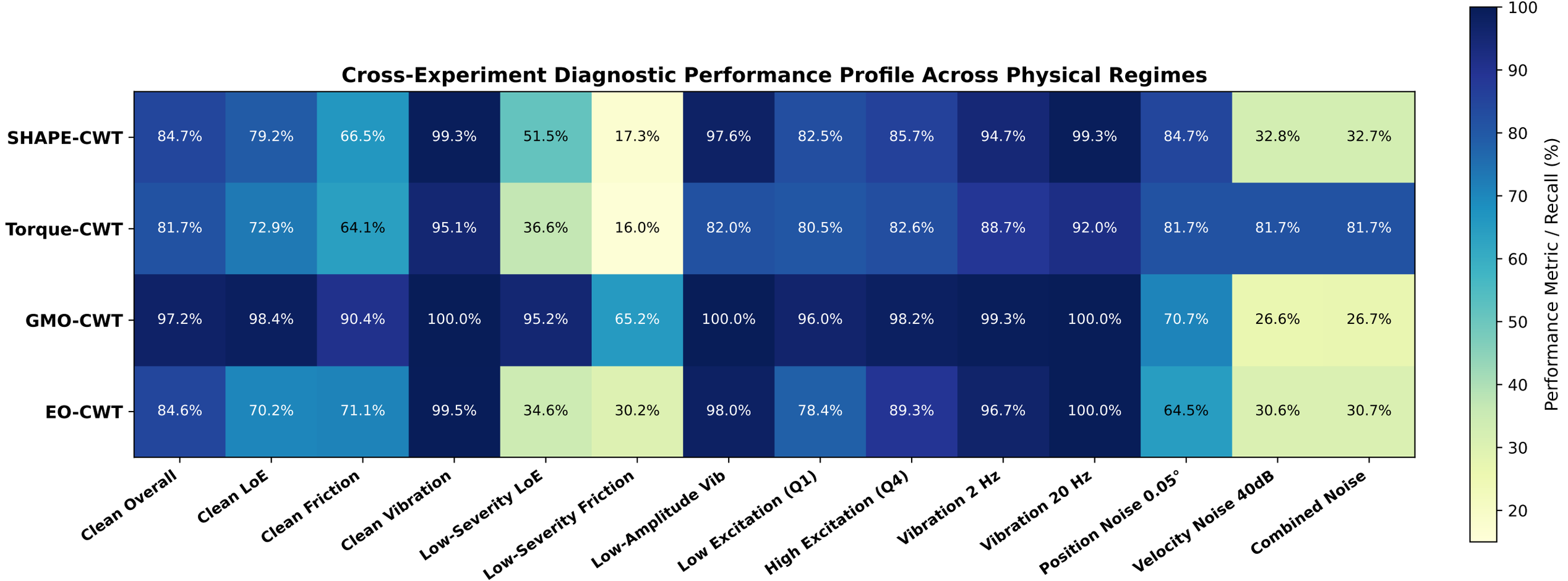

Cross-Experiment Diagnostic Performance Profile Across Physical Regimes
SHAPE-CWT
Torque-CWT
GMO-CWT
EO-CWT
84.7% 79.2% 66.5% 99.3% 51.5% 17.3% 97.6% 82.5% 85.7% 94.7% 99.3% 84.7% 32.8% 32.7%
81.7% 72.9% 64.1% 95.1% 36.6% 16.0% 82.0% 80.5% 82.6% 88.7% 92.0% 81.7% 81.7% 81.7%
97.2% 98.4% 90.4% 100.0% 95.2% 65.2% 100.0% 96.0% 98.2% 99.3% 100.0% 70.7% 26.6% 26.7%
84.6% 70.2% 71.1% 99.5% 34.6% 30.2% 98.0% 78.4% 89.3% 96.7% 100.0% 64.5% 30.6% 30.7%
Clean Overall
Clean LoE
Clean Friction
Clean Vibration
Low-Severity LoE
Low-Severity Friction
Low-Amplitude Vib
Low Excitation (Q1)
High Excitation (Q4)
Vibration 2 Hz
Vibration 20 Hz
Position Noise 0.05°
Velocity Noise 40dB
Combined Noise
Performance Metric / Recall (%)
100
90
80
70
60
50
40
30
20